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Inference of the Trend in a Partially Linear Model with Locally Stationary Regressors

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In this article, we construct the uniform confidence band (UCB) of nonparametric trend in a partially linear model with locally stationary regressors. A two-stage semiparametric regression is employed to estimate the trend function. Based on this estimate, we develop an invariance principle to construct the UCB of the trend function. The proposed methodology is used to estimate the Non-Accelerating Inflation Rate of Unemployment (NAIRU) in the Phillips Curve and to perform inference of the parameter based on its UCB. The empirical results strongly suggest that the U.S. NAIRU is time-varying.

Keywords Invariane principle; Local-stationarity; Nonparametric trend; Partially linear model; Phillips curve; Semiparametric regression; Time-varying NAIRU; Uniform confidence band.

JEL Classification C12; C13; C14.

1. INTRODUCTION

The purpose in this article is to construct the *uniform confidence band* (UCB) of the trend in a partially linear model with locally stationary regressors. A partially linear model has been popular in the semiparametric econometrics literature due to its flexibility to incorporate parametric and nonparametric components together. For an excellent treatment of this literature, we refer to Härdle et al. (2000). The partially linear model of our interest has the framework

$$y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \mu(i/n) + \epsilon_i, \tag{1}$$

where $i = 1, 2, \ldots, n$.[1] Note that y_i is the observed response variable and that ϵ_i is a stationary random error satisfying $\mathbb{E}(\epsilon_i|\mathbf{x}_i) = 0$. Here $\mathbf{x}_i = (x_{i1}, x_{i2}, \ldots, x_{ip})^\top$ is a $(p \times 1)$ random vector of observed regressors. We let $\mathbf{x}_i$ be *locally stationary* (Dahlhaus,

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[1]We use index i for time. The traditional time index t is reserved for continuous time in $[0, 1]$.

1997; Kim et al., 2010; Koo and Linton, 2012; Vogt, 2012), which is a mild form of nonstationarity (see Assumption 2). A $(p \times 1)$ vector $\boldsymbol{\beta}$ and a scalar $\mu(\cdot)$ are unobserved. The difference is that $\boldsymbol{\beta}$ is a vector of fixed parameters, while $\mu(\cdot)$ is a deterministic trend. Gao and Hawthorne (2006) studies (1) with independent and trend-stationary $\mathbf{x}_i$. We extend it to the case of dependent and locally stationary $\mathbf{x}_i$. For more on the assumptions for (1), we refer to Section 2. In this article, our interest is in trend $\mu(\cdot)$. We estimate and carry out inference of parameter $\mu(\cdot)$ in (1).

Traditionally, the literature on semiparametric models has focused on estimating $\boldsymbol{\beta}$ rather than $\mu(\cdot)$. Consistent estimators of $\boldsymbol{\beta}$ have been proposed by Robinson (1988) and Yatchew (1997), among others. Despite this tradition, inference of $\mu(\cdot)$ based on its UCB could be useful and valuable, given the importance of a time trend in an economic variable. One useful application of the framework (1) is to the *Phillips Curve*, where the long-term trend in unemployment rate is usually called the Non-Accelerating Inflation Rate of Unemployment (NAIRU). In macroeconomics, NAIRU plays an important role because this structural parameter allows us to determine what the current status of the economy is in business cycles. If the current unemployment rate is below the NAIRU, the economy is believed to be in an economic expansion. Otherwise, it is in a recession. Staiger et al. (1996, 1997) introduce the Phillips Curve with a *fixed* slope coefficient but a *time-varying* NAIRU, which is an example of our semiparametric model (1). By constructing its UCB, we can conduct model validation for the important NAIRU parameter.

To construct the UCB of the trend $\mu(\cdot)$ in (1), we first propose *two-stage semiparametric* estimation of $\mu(\cdot)$. We assume that $\mu(\cdot)$ is a nonparametric function that changes *smoothly* in continuous time. Specifically, we assume that the second derivative of $\mu(\cdot)$ is continuous over a fixed time domain $[0, 1]$. That is, $\mu(\cdot) \in \mathscr{C}^2[0, 1]$. A combination of the differencing estimation (Yatchew, 1997) and the local-linear regression (Cleveland, 1979; Fan and Gijbels, 1996) is utilized to estimate $\mu(\cdot)$ in (1). We apply the *strong invariance principle* (Wu, 2007) to the two-stage semiparametric estimate of $\mu(\cdot)$ to construct the UCB of $\mu(\cdot)$. A detailed description of these procedures is provided in Sections 3 and 4.

In fact, it is common in economics to assume a smoothly time-varying process on structural parameters such as $\mu(\cdot)$ in (1). For example, many economists believe that the movement in the NAIRU of the Phillips Curve is *smooth* (Ball and Mankiw, 2002; Gordon, 1997, 1998; Staiger et al., 1996, 1997). They impose parametric structures on time-varying NAIRU to estimate it. Cai (2007) introduces the slowly time-varying *beta-coefficient* to the traditional Capital Asset Pricing Model (CAPM) and estimates the coefficient based on the local-linear approach. González and Teräsvirta (2008) and González et al. (2009) assume a smoothly time-varying parametric function for *core inflation* and use their parametric models to forecast the U.S. inflation series. Kim (2014) considers the Consumption-based CAPM (C-CAPM) where the *risk aversion* parameter is assumed to evolve slowly in time. He proposes a consistent estimator of time-varying risk aversion and studies the parameter's cyclical behavior based on the proposed estimate.

Kim et al. (2010) introduce a partial equilibrium model with smoothly time-varying price/income elasticities for the U.S. durable goods market. They provide the empirical evidence that the demand elasticities in their model indeed change slowly in time. Given these implications, we assume that trend $\mu(\cdot)$ in (1) evolves smoothly over time.

Our main contribution is that we perform inference of $\mu(\cdot)$ in (1) based on its UCB when the regressors are *locally stationary*. Wu and Zhao (2007) consider a random process that is decomposed into a deterministic trend and a random error, and suggest how to perform inference for the trend. Given the simple structure of their model, the scope of applications is rather limited. Gao and Hawthorne (2006) work on a more general framework, where regressors are added to the process. However, they assume that the regressors are independent and trend-stationary. We extend Wu and Zhao (2007) and Gao and Hawthorne (2006) by adding regressors that are both locally stationary and temporally dependent. The inference for the trend is performed by constructing its UCB. In contrast to the traditional point-wise confidence intervals, the UCB can be used to test for the overall shape of an unknown function. By checking whether a parametric function is *uniformly* contained by the UCB, one can determine whether the function is suitable for the trend or not.

Secondly, we develop the *invariance principle* to construct the UCB of $\mu(\cdot)$ in (1), which is the main *theoretical* contribution of the article. Typically, the construction of UCB utilizes the asymptotic Gumbel (or extreme-value) distribution. As a result, the UCB based on the Gumbel distribution involves unknown centering and scaling parameters that need to be handled (see Remark 7). In contrast, the invariance principle allows us to utilize the supremum of a partial sum involving independent and identically distributed (IID) standard normals. This simulation-based approach to construct the UCB of trend is possibly more convenient than the one based on the Gumbel, since we do not have to deal with the centering and scaling parameters under our approach. A similar idea is discussed rather informally for the purely nonparametric model in Wu and Zhao (2007). We extend the idea to the more general/practical *semiparametric* case, and formally prove the principle in Theorem 2.

The organization of the article is as follows. Section 2 discusses the assumptions for our asymptotic results. Section 3 introduces the two-stage semiparametric regression to estimate the trend $\mu(\cdot)$ in (1) over a compact time domain. The first-stage parametric and the second-stage nonparametric estimations are carefully explained. Section 4 describes how to construct the UCB of $\mu(\cdot)$. To that end, we show how to derive the *invariance principle* (i.e., Theorem 2) involving the proposed estimator. Detailed steps to construct the UCB and a simulation study are also provided. As an application, we estimate and perform inference of the NAIRU in the Phillips Curve in Section 5. The empirical results are explained in detail. Section 6 concludes the article and discusses related future research. The mathematical proofs regarding the construction of UCB are relegated to the Appendix.

2. ASSUMPTIONS

We first introduce some notations that will be used throughout this study. For any vector $\mathbf{v} = (v_1, v_2, \ldots, v_p) \in \mathbb{R}^p$, we let $|\mathbf{v}| = \left(\sum_{i=1}^p v_i^2\right)^{1/2}$. For any random vector $\mathbf{V}$, we write $\mathbf{V} \in \mathscr{L}^q$ ($q > 0$) if $\|\mathbf{V}\|_q = \left[\mathbb{E}\left(|\mathbf{V}|^q\right)\right]^{1/q} < \infty$. In particular, $\|\mathbf{V}\| = \|\mathbf{V}\|_2$. We denote $\mathbf{L} : [0,1] \times \mathbb{R}^\infty \mapsto \mathbb{R}^p$ as a measurable function such that $\mathbf{L}(t, \mathscr{F}_i)$ is a properly defined $(p \times 1)$ random vector for all $t \in [0,1]$, where $\mathscr{F}_i = (\cdots, \eta_{i-1}, \eta_i)$ with IID random errors $\{\eta_i\}_{i\in\mathbb{Z}}$. Define the *physical dependence measure* (Wu, 2005) for $\mathbf{L}(t, \mathscr{F}_i)$ as

$$\delta_q(\mathbf{L}, k) = \sup_{t\in[0,1]} \|\mathbf{L}(t, \mathscr{F}_k) - \mathbf{L}(t, \mathscr{F}_k^*)\|_q, \tag{2}$$

where $\mathscr{F}_i^* = (\cdots, \eta_0^*, \ldots, \eta_{i-1}, \eta_i)$ is a coupled process with η_0^* being an IID copy of η_0. For discussion on this dependence measure, we refer to Remark 4. For a class of stochastic processes $\{\mathbf{L}(t, \mathscr{F}_i)\}_{i\in\mathbb{Z}}$, we say that the process is $\mathscr{L}^q$ *stochastic Lipschitz-continuous* over $[0,1]$ if

$$\sup_{0\le t_1<t_2\le 1} \frac{\|\mathbf{L}(t_2, \mathscr{F}_0) - \mathbf{L}(t_1, \mathscr{F}_0)\|_q}{|t_2 - t_1|} < \infty. \tag{3}$$

We denote a collection of such systems by Lip_q. In addition, we write $a_n \asymp b_n$ if $|a_n/b_n|$ is bounded away from 0 and ∞ for all large n. Given these notations, we introduce the following assumptions that will be used throughout the article. Given model (1), we have the following assumptions.

Assumption 1. *The regressors are nonstationary such that*

$$\mathbf{x}_i = \mathbf{G}(i/n,\ \mathscr{U}_i), \quad i = 1, 2, \ldots, n, \tag{4}$$

where $\mathscr{U}_i = (\cdots, u_{i-1}, u_i)$ with $\{u_i\}_{i\in\mathbb{Z}}$ being a set of IID random elements. Here $\mathbf{G} := (G_1, G_2, \ldots, G_p)^\top : [0,1] \times \mathbb{R}^\infty \to \mathbb{R}^p$ is a measurable function such that $\mathbf{G}(t,\ \mathscr{U}_i)$ is well defined for each $t \in [0,1]$.

Assumption 2. *Let $\mathbf{G}(t,\ \mathscr{U}_i) \in Lip_2$ and $\sup_{0\le t\le 1} \|\mathbf{G}(t,\ \mathscr{U}_i)\|_4 < \infty$.*

Assumption 3. *Let $\mathbf{G}_d(i/n,\ \mathscr{U}_i) = \mathbf{G}(i/n,\ \mathscr{U}_i) - \mathbf{G}((i-1)/n,\ \mathscr{U}_{i-1})$. Define the $p \times p$ matrix $M_d(t) := \mathbb{E}[\mathbf{G}_d(t,\ \mathscr{U}_0) \cdot \mathbf{G}_d(t,\ \mathscr{U}_0)^\top]$ such that the smallest eigenvalue of $M_d(t)$ is bounded away from 0 on $t \in [0,1]$.*

Assumption 4. *Let ϵ_i be a stationary random error with $\mathbb{E}(\epsilon_i|\mathbf{x}_i) = 0$ and $\mathbb{E}|\epsilon_i|^r < \infty$, $r > 3$ such that*

$$\epsilon_i = H(\mathscr{V}_i), \quad i = 1, 2, \ldots, n, \tag{5}$$

where $\mathcal{V}_i = (\cdots, v_{i-1}, v_i)$ *with* $\{v_i\}_{i\in\mathbb{Z}}$ *being a set of IID random elements. Here* $H : \mathbb{R}^\infty \to \mathbb{R}$ *is a well-defined measurable function. Furthermore, let*

$$\sum_{j=1}^{\infty} j\|\epsilon_j - \epsilon_j^*\|_r < \infty, \tag{6}$$

where $\epsilon_j^* = H(\cdots, v_0^*, \ldots, v_{j-1}, v_j)$ *with* v_0^* *being an IID copy of* v_0.

Assumption 5. *Let* $\zeta_i = (u_i, v_i)^\top$ *and* $\mathcal{I}_i = (\cdots, \zeta_{i-2}, \zeta_{i-1}, \zeta_i)$. *Define* $\mathbf{U}(t, \mathcal{I}_i) = \mathbf{G}(t, \mathcal{U}_i)H(\mathcal{V}_i)$. *Then,* $\sum_{k=0}^{\infty}[\delta_4(\mathbf{G}, k) + \delta_2(\mathbf{U}, k)] < \infty$.

Assumption 6. *Let* $\mu(\cdot)$ *be twice continuously differentiable over* $[0, 1]$. *In other words,* $\mu(\cdot) \in \mathscr{C}^2[0, 1]$.

Assumption 7. *A kernel* $K(\cdot)$ *is bounded, positive, and symmetric, and has a compact support* $[-1,\ 1]$, *and is Lipschitz-continuously differentiable.*

Assumption 8. *A bandwidth* b_n *satisfies the condition*

$$nb_n^5 + \frac{(\log(n))^2}{n^{1-2/q}b_n} = o(1), \tag{7}$$

where $q = \min(r, 4)$ *and* r *is introduced by Assumption 4.*

We employ these assumptions to prove the lemmas and theorems in our article. For detail on the proofs, we refer to the Appendix. These assumptions are noteworthy, and we have the following important remarks.

Remark 1. Assumption 1 allows regressors $\mathbf{x}_i$ in model (1) to be *nonstationary* since their moments are time-varying. It is often reasonable to assume that the characteristics of model regressors vary in time. In particular, we let the time variation in these characteristics be *smooth*, rather than abrupt. Note also that $\mathbf{x}_i$ is *dependent* due to the cumulative IID random elements.

Remark 2. Assumption 2 ensures that model regressors $\mathbf{x}_i$ are *locally stationary*, which is a mild form of nonstationarity. That is, if one observes locally stationary variables in a relatively short time span, they are approximately stationary. Since many economic variables including unemployment rates are possibly locally stationary processes (see Fig. 1), we can make our model specification more general by introducing this assumption. For more on the local-stationarity, we refer to Priestley (1965), Dahlhaus (1997), Mallat et al. (1998), Ombao et al. (2005), Kim et al. (2010), Koo and Linton (2012) and Vogt (2012). Assumption 3 prevents the asymptotic *multicollinearity* of regressors.

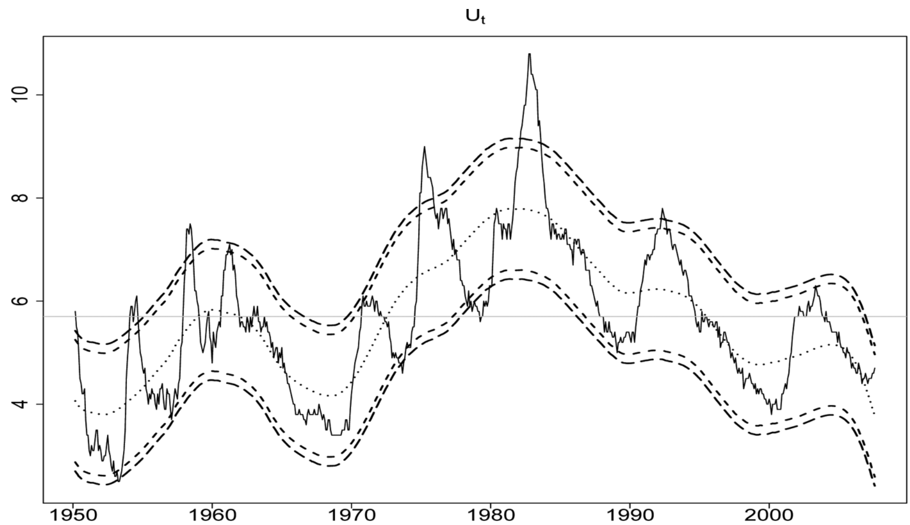

FIGURE 1 Time-varying NAIRU for the U.S. economy (1950–2007): The *short-dashed* curves are 95% uniform confidence band (UCB) of NAIRU, while the *long-dashed* ones are 99% UCB. The widths of 95% and 99% UCBs are 2.38% and 2.73%, respectively. The *dotted* curve in the middle of the UCBs is the two-stage estimate of time-varying NAIRU. For the local linear regression, we use an Epanechnikov kernel. The GCV chooses $b_n = 0.11$. The estimate of NAIRU and the UCB are placed over the U.S. monthly unemployment rates (solid curve). The horizontal line is the estimate of *fixed* NAIRU, 5.70%.

Remark 3. Assumption 4 allows error term ϵ_i with bounded rth moments to be dependent and possibly correlated. The dependence structure for ϵ_i defined by (5) is flexible and general in that function $H(\cdot)$ is *not* specified. Note that the cumulative dependence structure (5) and the condition (6) ensure that the strong invariance principle (Wu, 2007) applies to our model framework. Examples of random process ϵ_i include a stationary ARCH process (Engle, 1982). Assumption 4 also ensures the exogeneity of regressors $\mathbf{x}_i$.

Remark 4. Assumption 5 ensures *short-range dependence* among the variables in our model. The interpretation is that the cumulative effect of a single error on all future values is bounded. The measure of dependence used here is the *physical dependence measure* (Wu, 2005) based on causal processes. This measure is known to be useful for characterizing dependence in *nonlinear* time series models (Wu, 2005, 2007; Wu and Zhao, 2007; Liu and Wu, 2010; Kim, 2014; Kim et al., 2010). The boundedness condition in Assumption 5 implies that the physical dependence measure becomes zero as the lag k goes to infinity, which is a notion similar to the traditional *strong-mixing* condition. While the strong mixing involves the computation of probabilistic dependence measures, the physical dependence here utilizes (2). For more on the comparison of these two measures, we refer to Wu (2005).

Remark 5. Assumption 6 guarantees that $\mu(\cdot)$ changes *smoothly* in time. The second-order continuity is required for the consistency of the *local-linear estimate* (see Section 3) for the first-order derivative.

Remark 6. Assumption 7 allows popular kernel functions such as the Epanechnikov kernel. Assumption 8 provides the convergence order of bandwidth b_n.

3. TWO-STAGE ESTIMATION OF TREND

3.1. First-Stage Parametric Estimation

Since the partially linear model (1) has both parameter $\boldsymbol{\beta}$ and nonparametric function $\mu(\cdot)$, it is natural to employ a *two-step* method to estimate both unknowns. A similar two-step method for a different framework is considered in Kim (2014), among many others. We first estimate $\boldsymbol{\beta}$ based on the first-differenced version of model (1)

$$\Delta y_i = \Delta \mathbf{x}_i^{\top} \boldsymbol{\beta} + \left[\mu\left(\frac{i}{n}\right) - \mu\left(\frac{i-1}{n}\right) \right] + \Delta \epsilon_i, \tag{8}$$

where $\Delta y_i = y_i - y_{i-1}$, $\Delta \mathbf{x}_i = \mathbf{x}_i - \mathbf{x}_{i-1}$, and $\Delta \epsilon_i = \epsilon_i - \epsilon_{i-1}$, respectively. Since $\mu(\cdot) \in \mathscr{C}^2[0,1]$, for all $i = 1, 2, \ldots, n$, we have the result

$$\left| \mu\left(\frac{i}{n}\right) - \mu\left(\frac{i-1}{n}\right) \right| = O\left(\frac{1}{n}\right) \tag{9}$$

which means that the left-hand side of (9) disappears quickly as sample size n increases. Then, we can rewrite (8) as

$$\Delta y_i = \Delta \mathbf{x}_i^{\top} \boldsymbol{\beta} + O(1/n) + \Delta \epsilon_i. \tag{10}$$

The first-stage estimation of unknown parameter $\boldsymbol{\beta}$ in (1) is based on Eq. (10). The idea of differencing is introduced by Yatchew (1997). In this study, we employ the least squares estimation method to estimate $\boldsymbol{\beta}$ in (1) and (8), treating $\Delta \mathbf{x}_i$ as the regressor and Δy_i as the response variable. That is, we propose the *differencing estimator*

$$\hat{\boldsymbol{\beta}}_D = \left(\sum_{i=1}^{n} \Delta \mathbf{x}_i \Delta \mathbf{x}_i^{\top} \right)^{-1} \sum_{i=1}^{n} \Delta \mathbf{x}_i \Delta y_i, \tag{11}$$

where $\hat{\boldsymbol{\beta}}_D$ is based on the differenced data. The weak consistency and limiting distribution for the differencing estimator are given by Lemma 3 and by Remark 8 in Appendix.

Yatchew (1997) proposed the differencing estimate under IID regressors and error. Here the regressors are *locally stationary*, and both the regressors and error of (1) are *dependent*. Robinson (1988) introduced his kernel-based estimation of $\boldsymbol{\beta}$ in (1). Both the differencing estimate and the Robinson's estimate are $\sqrt{n}$-consistent. However, we prefer to employ the differecing method in this work, because the the Robinson's method involves separate nonparametric regressions for each parametric variable and for the dependent variable. On the other hand, the differencing estimation used here avoids the preliminary nonparametric regressions required for the Robinson's method. Given the locally stationary regressors of (1), the smoothness condition on $\mu(\cdot)$ ensures that the differencing estimate $\hat{\boldsymbol{\beta}}_D$ is $\sqrt{n}$-consistent, as shown by Lemma 3. Alternatively, one can employ the Robinson's estimate for the first-stage estimation.

3.2. Second-stage Nonparametric Estimation

Given the differencing estimate of $\boldsymbol{\beta}$ in (11), we are ready to estimate the trend $\mu(t)$. We first rewrite the partially linear model (1), and obtain

$$y_i - \mathbf{x}_i^\top \boldsymbol{\beta} = \mu(i/n) + \epsilon_i, \tag{12}$$

where ϵ_i is a stationary random error satisfying $\mathbb{E}|\epsilon_i|^r < \infty$, $r > 3$. Given that $\mu(t)$ changes *smoothly* over $t \in [0, 1]$, we employ the *local linear regression* to estimate the trend function $\mu(\cdot)$. Among many available nonparametric methods, local polynomial regression (Cleveland, 1979; Fan and Gijbels, 1996) is frequently used due to its simple form, ease of computation and analytical tractability. In contrast to other popular nonparametric methods such as the Nadaraya–Watson estimator (i.e., local constant estimator), a local linear estimator does suppress the well-known *boundary problem* and achieve nearly optimal statistical efficiency (Fan and Gijbels, 1996). Thus, we shall use a local linear estimate of $\mu(\cdot)$ in this study. The local linear regression gives

$$(\hat{\mu}(t), \hat{\mu}'(t)) := \operatorname{argmin}_{(\eta_0,\eta_1)} \sum_{i=1}^{n} \left[y_i - \mathbf{x}_i^\top \boldsymbol{\beta} - \eta_0 - \eta_1(t_i - t)\right]^2 K\left(\frac{t_i - t}{b_n}\right), \tag{13}$$

where $t_i = i/n$ and $i = 1, \ldots, n$. Here $K(\cdot)$ is a kernel function and b_n is a bandwidth. A popular choice is the Epanechnikov kernel $K(x) = 3\max(1 - x^2, 0)/4$. The bandwidth b_n can be viewed as the size of the neighborhood on which $\mu(\cdot)$ is estimated. If b_n is large, then the estimation is based on a large neighborhood and the estimated $\hat{\mu}(t)$ will have a small variance and a large bias. Analogously, if b_n is small, $\hat{\mu}(t)$ will have a large variance and a small bias. Popular ways to select the bandwidth in practice include the generalized cross-validation (GCV) (Craven and Wahba, 1979). Here $\hat{\mu}(\cdot)$ is called the *local linear estimate* of $\mu(\cdot)$. As a by-product, (13) also gives an estimate of its derivative $\mu'(\cdot)$. Since (13) is essentially a weighted least squares estimate, we can write the solution of (13) for $\mu(\cdot)$ as

$$\hat{\mu}(t) = \sum_{i=1}^{n} w_n(t, i) \left(y_i - \mathbf{x}_i^\top \boldsymbol{\beta}\right), \tag{14}$$

where $w_n(t,i) = K\left(\frac{t-i/n}{b_n}\right)\frac{S_2(t)-(t-i/n)S_1(t)}{S_2(t)S_0(t)-S_1^2(t)}$ with $S_j(t) = \sum_{i=1}^{n}(t-i/n)^j K\left(\frac{t-i/n}{b_n}\right), j = 0,1,2.$ The time domain of t is fixed over $t \in [0,1]$ and $w_n(t,i)$ is the weight given to each observation. Note here that estimator (14) is *infeasible* because $\boldsymbol{\beta}$ is *unknown*. Hence, we propose the following *feasible* estimator:

$$\tilde{\mu}(t) = \sum_{i=1}^{n} w_n(t,i)\left(y_i - \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}_D\right), \tag{15}$$

where $\hat{\boldsymbol{\beta}}_D$ is the differencing estimate of $\boldsymbol{\beta}$ in (11). We call $\tilde{\mu}(t)$ the two-stage semiparametric estimator of $\mu(\cdot)$ since the first-stage parametric estimate $\hat{\boldsymbol{\beta}}_D$ is embedded in the second-stage nonparametric estimate of $\mu(\cdot)$. The consistency of this two-stage semiparametric estimator (15) is given by the following theorem.

Theorem 1 (Consistency). *Under Assumptions 1–8, for any fixed $t \in (0,1)$,*

$$\left|\tilde{\mu}(t) - \mu(t)\right| = O_{\mathbb{P}}\left(b_n^2 + \frac{1}{\sqrt{nb_n}} + \frac{1}{\sqrt{n}}\right), \tag{16}$$

where $\tilde{\mu}(t)$ is the two-stage semiparametric estimator for $\mu(t)$ defined by (15).

The proof of Theorem 1 is provided in the Appendix. By Assumption 8, the consistency of $\tilde{\mu}(t)$ to $\mu(t)$ follows easily. Based on this consistent semiparametric estimate, we shall construct the uniform confidence band of $\mu(\cdot)$ such that we can carry out inference of the unknown parameter function.

4. UNIFORM CONFIDENCE BAND (UCB)

In order to construct the asymptotic UCB of trend $\mu(t)$ over $t \in [0,1]$ with the confidence level $100(1-\alpha)\%, \alpha \in (0,1)$, we need to find the following two functions $f_n(\cdot)$ and $g_n(\cdot)$ based on data:

$$\lim_{n\to\infty} \mathbb{P}\{f_n(t) \le \mu(t) \le g_n(t) \text{ for all } t \in \mathcal{T}\} = 1-\alpha, \tag{17}$$

where $\mathcal{T} = [0,1]$. The purpose of constructing the UCB in (17) is to test whether the trend $\mu(\cdot)$ takes a certain parametric form. That is, using (17), we are able to test the null hypothesis $H_0 : \mu(\cdot) = \mu_\theta(\cdot)$, where $\theta \in \Theta$ and Θ is a parameter space. For example, in order to test $H_0 : \mu(t) = \theta_0 + \theta_1 t + \theta_2 t^2$, one can simply check whether $f_n(t) \le \hat{\theta}_0 + \hat{\theta}_1 t + \hat{\theta}_2 t^2 \le g_n(t)$ holds for *all* $t \in \mathcal{T}$. Here $\hat{\theta}_i$ is the least squares estimate of θ_i, $i = 0,1,2$. If the condition does hold for *all* $t \in \mathcal{T}$, then we *fail to reject* the null hypothesis at level α.

The advantage of the UCB over the traditional point-wise bands is that the UCB can be used for inference of an unknown function, such as a trend. In addition, the UCB is a more *conservative* confidence band than the point-wise ones in that the UCB is *wider* than its point-wise counterpart. Thus, any test results based on the UCB would be more *robust* than those under the point-wise ones. For these reasons, the UCB has recently gained more attention in econometrics and statistics literature. For example, Härdle and Song (2010) construct the UCB of conditional quantiles for the age-earning relation in the U.S. labor market. Liu and Wu (2010) construct the UCBs of the mean and volatility of the U.S. Treasury yield curve rates. Given these interesting results, we shall construct the UCB of the trend in the partially linear model (1), and carry out inference. To our knowledge, this is the first time that one performs a formal test on parametric specifications of the trend in model (1) based on its UCB.

4.1. Invariance Principle

In constructing the UCB for the trend $\mu(\cdot)$ in model (1), we shall employ a very useful tool in asymptotic theories, called the *strong invariance principle*. The principle basically allows us to access asymptotic properties of a partial sum of stationary and ergodic random variables. Komlós et al. (1975) introduce their celebrated strong invariance principle for IID random variables. Wu (2007) extends the idea by developing his invariance principle for dependent random variables. Given the dependence among ϵ_i in (1), we shall employ the strong invariance principle in Wu (2007). Under (6), Wu (2007) shows that there is a richer probability space such that $\max_{i\le n}\left|\sum_{k=1}^{i}\epsilon_k-\sigma_\epsilon\mathbb{B}(i)\right| = o_{\mathbb{A}\mathbb{S}}\left(n^{1/q}\log(n)\right)$. Here $\sigma_\epsilon^2 := \sum_{k\in\mathbb{Z}}\mathbb{E}(\epsilon_0\epsilon_k)$ is the long-run variance of ϵ_i in (1), and $\mathbb{B}(\cdot)$ is the standard Brownian motion. By applying this result, we obtain the main theoretic result of this article.

Theorem 2 (Invariance Principle). *Recall that $\tilde{\mu}(t)$ is the two-stage semiparametric estimator of $\mu(t)$ from (15). Under Assumptions 1–8,*

$$\sqrt{nb_n}\sup_{0\le t\le 1}\left|\tilde{\mu}(t)-\mu(t)-\sigma_\epsilon\sum_{i=1}^{n}w_n(t,i)Z_i\right| = o_{\mathbb{P}}(1), \tag{18}$$

where $\sigma_\epsilon^2 = \sum_{k\in\mathbb{Z}}\mathbb{E}(\epsilon_0\epsilon_k)$ and Z_i is an IID standard normal random variable.

The proof of Theorem 2 is given in the Appendix. The theorem implies that the uniform convergence order of the difference between $\tilde{\mu}(t)-\mu(t)$ and $\sigma_\epsilon\sum_{i=1}^{n}w_n(t,i)Z_i$ is $o_{\mathbb{P}}\left(1/\sqrt{nb_n}\right)$. In other words, the distribution of $\sup_{0\le t\le 1}\left|\tilde{\mu}(t)-\mu(t)\right|$ can be well-approximated by that of $\sigma_\epsilon\sup_{0\le t\le 1}\left|\sum_{i=1}^{n}w_n(t,i)Z_i\right|$. Hence we construct the UCB of $\mu(\cdot)$ based on the estimate of $\mu(\cdot)$, the estimate of σ_ϵ and the sampling distribution of $\sup_{0\le t\le 1}\left|\sum_{i=1}^{n}w_n(t,i)Z_i\right|$ under Theorem 2.

Interestingly, the limiting distribution of normalized $\sup_{0\le t\le 1}\left|\tilde{\mu}(t)-\mu(t)\right|$ is a Gumbel distribution (Bickel and Rosenblatt, 1973; Johnston, 1982; Hall, 1991; Eubank and Speckman, 1993; Fan and Zhang, 2000; Wu and Zhao, 2007). See Remark 7 for the detail.

For this reason, the UCB of $\mu(\cdot)$ is typically constructed based on the asymptotic Gumbel distribution. Thus, Theorem 2 provides us with an alternative way to construct the UCB of $\mu(\cdot)$ under (18) without using the asymptotic Gumbel distribution. It is also noteworthy that the UCB construction under (18) requires the estimation of the long-run variance σ_ϵ^2 only, while the UCB construction based on the asymptotic Gumbel distribution requires the estimation of unknown centering and scaling parameters as well (see Remark 7).

4.2. Construction of the UCB

One of the conditions required to ensure Theorem 2 is Assumption 8, the *undersmoothing* condition. In practice, this condition could cause a problem in selecting the optimal bandwidth for the proposed two-stage estimator. One way to solve this potential problem is *bias-correction*. We consider the *jackknife-type* bias-corrected estimate (Härdle, 1986)

$$\tilde{\mu}^*(t) = \sum_{i=1}^{n} w^*(t,i)\left(y_i - \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}_D\right), \tag{19}$$

where $w^*(t,i) := 2w_n(t,i) - w_{b\sqrt{2}}(t,i)$. Here $w_{b\sqrt{2}}(t,i)$ is the local-linear weight $w_n(t,i)$ in (14) with $b_n\sqrt{2}$ instead of b_n. A careful check on the proof of Theorem 2 shows that, under $nb_n^7 + \frac{(\log(n))^2}{n^{1-2/q}b_n} = o(1)$,

$$\sqrt{nb_n}\sup_{0\le t\le 1}\left|\tilde{\mu}^*(t) - \mu(t) - \sigma_\epsilon \sum_{i=1}^{n} w^*(t,i)Z_i\right| = o_{\mathbb{P}}(1). \tag{20}$$

In contrast to Assumption 8, the bandwidth condition $nb_n^7 + \frac{(\log(n))^2}{n^{1-2/q}b_n} = o(1)$ for (20) includes the mean-squared-error optimal bandwidth $b_n \asymp n^{-1/5}$. Thus, using popular bandwidth selection criteria, such as the GCV, is justified. Based on (20), we suggest the following procedures to construct the 95% UCB of $\mu(\cdot)$ in (1):

(i) Obtain the differencing estimate $\hat{\boldsymbol{\beta}}_D$ in (11);

(ii) (Bandwidth selection) Consider $\tilde{\mu}^*(t)$ in (19) under some b_n and the Epanechnikov kernel. Then, the fitted values would be $\hat{y}_i(b_n) = \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}_D + \tilde{\mu}^*(i/n), i = 1, \ldots, n$. Note here that $\hat{y}_i(b_n)$ depends on b_n due to $\tilde{\mu}^*(\cdot)$. To pick up the *optimal* bandwidth, we consider

$$M(b_n) = H(b_n)Y,$$

where $M(b_n) := \left(\hat{y}_1(b_n), \hat{y}_2(b_n), \ldots, \hat{y}_n(b_n)\right)^\top$, $Y := (y_1, \ldots, y_n)^\top$, and $H(b_n)$ is a $(n \times n)$ smoothing matrix that depends on b_n. The GCV criterion chooses the optimal bandwidth b_n^{opt} that minimizes

$$b_n^{opt} := \operatorname{argmin}_{b_n} \frac{n^{-1}\sum_{i=1}^n \left(y_i - \hat{y}_i(b_n)\right)^2}{\{1 - \mathrm{trace}[H(b_n)]/n\}^2},$$

where the numerator represents the goodness-of-fit and the denominator can be viewed as the model's degrees of freedom (Kim et al., 2010). We obtain (19) using b_n^{opt}.

(iii) Compute $\sup_{0\le t\le 1}\left|\sum_{i=1}^n w^*(t,i)Z_i\right|$, where $\{Z_i\}$ are generated IID standard normals.

(iv) Repeat (iii), say 1,000 times. We obtain the 95th quantile of the sampling distribution of $\sup_{0\le t\le 1}\left|\sum_{i=1}^n w^*(t,i)Z_i\right|$, and denote it as $\hat{q}_{0.95}$.

(v) Estimate σ_ϵ using the following *variant* of the subseries variance estimator proposed by Carlstein (1986):

$$\hat{\sigma}_\epsilon^2 = \frac{1}{2(m-1)k_n}\sum_{i=1}^{m-1}\left[\sum_{j=1}^{k_n}\left(y_{j+ik_n} - y_{j+(i-1)k_n} - (\mathbf{x}_{j+ik_n} - \mathbf{x}_{j+(i-1)k_n})^\top \hat{\boldsymbol{\beta}}_D\right)\right]^2, \quad (21)$$

where k_n is the length of subseries and $m = [n/k_n]$ is the largest integer not exceeding n/k_n. Carlstein (1986) shows that the optimal length of subseries is $k_n \asymp n^{1/3}$. Hence, we let $k_n \asymp n^{1/3}$ here. For a finite sample, we choose $k_n = [n^{1/3}]$. The consistency of $\hat{\sigma}_\epsilon^2$ to the long-run variance σ_ϵ^2 is given by Lemma 5 in Appendix.

(vi) The 95% UCB of $\mu(t)$ is

$$[\tilde{\mu}^*(t) \pm \hat{\sigma}_\epsilon \hat{q}_{0.95}] \quad (22)$$

Remark 7. Alternatively, one can construct the UCB of $\mu(t)$ based on the asymptotic distribution of normalized $\sup_{0\le t\le 1}\left|\tilde{\mu}(t) - \mu(t)\right|$. Given the $\sqrt{n}$–consistency of $\hat{\boldsymbol{\beta}}_D$ under Lemma 3 of this article, a straightforward extension of Theorem 2 in Wu and Zhao (2007) for our semiparametric estimator $\tilde{\mu}(t)$ shows

$$\mathbb{P}\left(\sqrt{\frac{nb_n}{\sigma_\epsilon^2\lambda_K}}\sup_{0\le t\le 1}\left|\tilde{\mu}(t) - \mu(t) - b_n^2\phi_K\mu''(t)\right| - d_n \le \frac{u}{\sqrt{2\log\left(b_n^{-1}\right)}}\right) \to e^{-2e^{-u}}, \quad (23)$$

where $\lambda_K = \int_{\mathbb{R}} K^2(u)du$, $\phi_K = \int_{\mathbb{R}} u^2K(u)du/2$, and d_n is a centering parameter under the Epanechnikov kernel with its bounded support

$$d_n = \sqrt{2\log\left(b_n^{-1}\right)} + \frac{1}{\sqrt{2\log\left(b_n^{-1}\right)}}\log\left(\frac{1}{\pi}\sqrt{\frac{1}{4\lambda_K}\int_{\mathbb{R}}(K'(u))^2du}\right).$$

The corresponding 95% UCB of $\mu(t)$ is

$$\left[\tilde{\mu}(t) - b_n^2 \phi_K \mu''(t) \ \pm \ \hat{\sigma}_\epsilon \sqrt{\frac{\lambda_K}{nb_n}} \left(d_n + \frac{q_{0.95}^G}{\sqrt{2\log\left(b_n^{-1}\right)}} \right) \right], \tag{24}$$

where $q_{0.95}^G$ is the 95th quantile of the Gumbel distribution in (23). To avoid the estimation of the bias term $b_n^2 \phi_K \mu''(t)$ in (24), one can instead use the biased-corrected estimate

$$\left[\tilde{\mu}^*(t) \pm \hat{\sigma}_\epsilon \sqrt{\frac{\lambda_K^*}{nb_n}} \left(d_n^* + \frac{q_{0.95}^G}{\sqrt{2\log\left(b_n^{-1}\right)}} \right) \right], \tag{25}$$

where λ_K^* and d_n^* represent λ_K and d_n under the higher-order kernel $K^*(u) = 2K(u) - K(u/\sqrt{2})/\sqrt{2}$ (Härdle, 1986) for the bias-correction, respectively.

4.3. Simulation

To illustrate the performance of the UCB based on Theorem 2, we carry out a simulation study. We consider the following model to compute the *coverage probability* of the proposed UCB in (22):

$$y_i = \beta x_i + \mu(i/n) + \epsilon_i, \quad i = 1, \dots, n, \tag{26}$$

where $\beta = 0.1$ and $\mu(t) = 0.15\sin(\pi t)$ for $0 \le t \le 1$. Under Assumptions 1–3, we propose the following structure on the regressor x_i:

$$x_i = \sum_{k=0}^{\infty} \gamma^k(i/n) u_{i-k} = \sum_{k=0}^{\infty} \left(\frac{1}{4} + \frac{i}{2n} \right)^k u_{i-k} \quad i = 1, \dots, n, \tag{27}$$

where u_i follows IID standard normal. Given Assumption 4, we consider two different specifications of ϵ_i, the threshold autoregressive (TAR) (Tong, 1990) and the autoregressive conditional heteroskedastic (ARCH) (Engle, 1982) processes

$$\begin{aligned} &\text{TAR}(1) : \epsilon_i = \theta \left| \epsilon_{i-1} \right| + v_i \sqrt{1 - \theta^2}, \quad i = 1, \dots, n, \\ &\text{ARCH}(1) : \epsilon_i = v_i \sqrt{\phi_0 + \phi_1 \epsilon_{i-1}^2}, \quad i = 1, \dots, n, \end{aligned}$$

where we let $\theta = 0.15$ for the TAR(1) and $\phi_0 = \phi_1 = 0.5$ for the ARCH(1). Here v_i also follows IID standard normal. We assume $\epsilon_0 = 0$. For a range of predetermined bandwidths (see Tables 1 and 2), we follow the procedures described in Section 4.2 to construct the 95% UCB of $\mu(\cdot)$ in (26). We repeat the process, say M times, for each one of the chosen bandwidths. Given M different UCBs, we count how many of those UCBs contain the true trend $\mu(t)$ in them, which gives us the desired coverage probabilities. Here we let $n = 500$ (sample size) and $M = 10,000$ (number of repetition). The simulation result for the TAR(1) error is reported in Table 1, while that for the ARCH(1) error is given by Table 2.

From Tables 1 and 2, we see that the coverage probabilities under various bandwidths are close to 0.95, which is the ideal coverage rate. For some bandwidths, we have the coverage probabilities that are almost identical to the nominal level 95%. However, as the bandwidth gets larger, the bias increases, which results in lower coverage probabilities for both the TAR(1) and ARCH(1) errors as we can observe from Tables 1 and 2, respectively.

In addition, we also report the simulation results based on the *GCV-chosen* bandwidths under various sample sizes. Table 3 shows the coverage probabilities of 95% UCB under the TAR(1) and ARCH(1) errors when the bandwidths are chosen by the GCV criterion introduced in Section 4.2. We consider seven different sample sizes from $n = 100$ to $n = 700$. As we can see from Table 3, the coverage probabilities get close to the ideal coverage probability 0.95 as the sample size gets large.

TABLE 1
Coverage Probabilities of 95% UCB Under the TAR(1) Error

b_n	0.11	0.12	0.13	0.14	0.15	0.16	0.17	0.18	0.19	0.20
prob	0.9642	0.9644	0.9584	0.9559	0.9426	0.9500	0.9521	0.9505	0.9438	0.9379

TABLE 2
Coverage Probabilities of 95% UCB Under the ARCH(1) Error

b_n	0.25	0.26	0.27	0.28	0.29	0.30	0.31	0.32	0.33	0.34
prob	0.9609	0.9633	0.9553	0.9584	0.9495	0.9446	0.9441	0.9278	0.9362	0.9251

TABLE 3
Coverage Probabilities of 95% UCB Under the GCV-Chosen b_n

n	100	200	300	400	500	600	700
$TAR(1)$	0.9016	0.8955	0.9516	0.9498	0.9580	0.9325	0.9571
$ARCH(1)$	0.8942	0.9115	0.9345	0.9457	0.9482	0.9396	0.9548

5. APPLICATION

5.1. Phillips Curve with Time-Varying NAIRU

One useful application of the framework (1) is to the inflation–unemployment tradeoff, which is known as the *Phillips Curve* in the macroeconomics literature. Specifically, we intend to construct the UCB of the NAIRU in the traditional Phillips Curve. One of the simple versions of the Phillips Curve in the literature is the following random-walk-type Phillips Curve (Staiger et al., 1996, 1997; Gordon, 1997, 1998; Fair, 2000; Ball and Mankiw, 2002):

$$\pi_i = \pi_{i-1} + \alpha(U_i - U_N) + \epsilon_i, \tag{28}$$

where π_i and U_i are inflation and unemployment rates at time $i = 1, \ldots, n$, respectively. Here ϵ_i is a mean zero random error at time i, often dubbed the *supply shock*. This equation has two fixed unknown parameters: α and U_N. Here U_N is the NAIRU that is also called the *natural rate* of unemployment. The parameter U_N embeds all shifts in the inflation-unemployment tradeoff. In that sense, the natural rate can be viewed as the unemployment rate that the economy reaches in the long run.

In principle, U_N can exhibit substantial *variation over time* (Gordon, 1997; Staiger et al., 1996, 1997; Ball and Mankiw, 2002). To implement this idea, we build on the traditional random-walk-type model (28) and propose the following Phillips Curve with *time-varying* NAIRU $U_N(\cdot)$

$$\pi_i = \pi_{i-1} + \alpha\left[U_i - U_N\left(\frac{i}{n}\right)\right] + \epsilon_i, \tag{29}$$

where $U_N(\cdot)$ is time-varying over its domain $[0, 1]$. Obviously, this model is a special case of the general framework (1).[2] One of the differences between our model (29) and the previous works in the time-varying-NAIRU literature (Ball and Mankiw, 2002; Gordon, 1997; Staiger et al., 1996, 1997) is that the domain of $U_N(\cdot)$ in (29) is bounded strictly between zero and one because of $i = 1, \ldots, n$. By using i/n instead of i for $U_N(\cdot)$, we can naturally restrict the domain of the nonparametric function to $[0, 1]$, which can help construct a consistent semiparametric estimator of $U_N(\cdot)$. As discussed previously, we assume that $U_N(\cdot)$ is second-order continuous over its fixed time domain $[0, 1]$. Hence, the methodology developed for (1) can be easily modified for (29), such that the time-varying

[2]From Figure 1, the U.S. unemployment rate appears to be a non-stationary (and possibly locally stationary) process. Thus, our assumption of locally stationary regressors for model (29) seems appealling.

NAIRU $U_N(\cdot)$ is estimated by the semiparametric estimator $\tilde{U}_N(t)$[3]

$$\tilde{U}_N(t) = \sum_{i=1}^{n} w^*(t,i)\left(U_i - \frac{\Delta\pi_i}{\hat{\alpha}_D}\right), \tag{30}$$

where $\Delta\pi_i = \pi_i - \pi_{i-1}$ and $\hat{\alpha}_D$ is the differencing estimate of the *fixed* parameter α in (29). The weight $w^*(t,i)$ is defined by (19). Given the procedures in Section 4.2, we can construct the UCB of time-varying NAIRU $U_N(\cdot)$ in (29).

5.2. Empirical Results

To construct the UCB of NAIRU for the U.S. economy, we use *monthly* unemployment rates and *monthly* Consumer Price Index (CPI) for the U.S. economy from March 1950 to September 2007. The data on both unemployment rate and CPI were obtained from the Global Insight Basic Economics database at the Wharton Research Data Services (WRDS) of the Wharton School, the University of Pennsylvania (http://wrds.wharton.upenn.edu). The estimated time-varying NAIRU and its 95% UCB are plotted along with the monthly unemployment data in Fig. 1. The same figure also presents the 99% UCB of the NAIRU parameter. Note that the 99% UCB is naturally wider than the 95% UCB.[4]

As we can observe from Fig. 1, there has been a substantial amount of fluctuation in the U.S. NAIRU between 1950 and 2007. Throughout the 1950s and the 1960s, the estimated NAIRU stays below 6%, although it rises during the 1950s and falls during the 1960s. It steadily increases during the 1970s. The peak in the estimated NAIRU comes in the early 1980s. Since then, it has declined gradually, which becomes most notable during the late 1990s. It reaches its trough around 2000, and then slightly *increases* after that. Ball and Mankiw (2002) and Staiger et al. (1997, 2001) report very similar findings to ours. One difference between our result and theirs is that we estimate NAIRU before 1960 and after 2000 as well, which means that our result is more extensive.

The contribution of our work is that we *test* parametric specifications of NAIRU. The traditional belief in macroeconomics is that the NAIRU is *constant*. Based on the 95% UCB of NAIRU (Fig. 1), we *reject*, at 5% level, the null hypothesis that the U.S. NAIRU is constant between 1950 and 2007 because the 95% UCB cannot contain any horizontal line in it (see Fig. 1). We *reject* the null even at 1% level since the 99% UCB cannot contain any horizontal line inside (see Fig. 1). Based on these findings, there seems to be a strong evidence against the hypothesis of time-invariant NAIRU. To our knowledge, this is the first article that rejects the uniform constancy of the U.S. NAIRU based on its UCB.

[3]We use an Epanechnikov kernel. The GCV chooses $b_n = 0.11$ for (30).

[4]The width of 95% UCB of time-varying NAIRU is 2.38% while that of 99% UCB is 2.73%.

Typically, inference of structural parameters is based on their *point-wise* confidence intervals. Staiger et al. (1997) employ this approach and report the 95% confidence intervals of U.S. NAIRU at three specific dates (first quarters of 1984, 1989, and 1994). From Table 1 (p. 39) of Staiger et al. (1997), the widths of CPI-based 95% confidence intervals of NAIRU at these three dates are 3.7%, 3.1%, and 3.7%, respectively. Since their results are point-wise, the width of interval varies from point to point. On the other hand, the width of our CPI-based 95% UCB of NAIRU is only 2.38%, while the width of 99% UCB is 2.73%.[5] Although our result is based on more observations, this could imply that our two-stage estimate of U.S. NAIRU along with its UCB is *informative* since our UCB has a smaller width than the point-wise ones reported in Table 1 of their article. From the same table in Staiger et al. (1997), our UCBs are either comparable to or narrower than the confidence intervals based on the different measures of U.S. inflation.

Furthermore, our estimation result shows that the Phillips Curve has a *negative* slope. The differencing estimate of α in (29) is $\hat{\alpha}_D = -0.7279$. The literature on the Phillips Curve with a *constant* NAIRU reports similar findings. Here we instead work with the Phillips Curve with a *time-varying* NAIRU. Our result seems to confirm the negative relationship between unemployment and inflation.

6. CONCLUSION

In this article, we construct the UCB of the trend in a partially linear model with locally stationary regressors. The main idea is to apply the invariance principle to a partial sum of model errors and to approximate the quantiles of the proposed statistic by using IID standard normals. The asymptotic justification of this approach is provided by Theorem 2. As an application, we employ the random-walk-type Phillips Curve and construct the UCB of NAIRU in the model. The results suggest that there is a substantial amount of variation in the U.S. NAIRU during the 1950–2007 period. The UCB of NAIRU at both 95% and 99% levels confirms the variation (Fig. 1). That is, the hypothesis of time-invariant NAIRU is *rejected* at both 5% and 1% levels by the UCB.

Regarding future research, this project suggests a couple of interesting topics for consideration. First, we can extend the current work to the case of *locally stationary error*. Thus, we can generalize the result found here. A rather straightforward extension of the current work will suffice. Second, we can investigate the *forecasting* ability of the proposed model with a time-varying trend. One of the main purposes of using the Phillips Curve in macroeconomics is to forecast inflation series. We can hopefully improve the forecasting ability of a partially linear model by adopting a time-varying trend in it. Further insight can be gained by extending the current work in these and other directions.

[5]The width of band is constant because the band is uniform and the model error is stationary.

7. APPENDIX

Recall that Assumption 3 defines the first-differenced regressors by

$$\Delta \mathbf{x}_i = \mathbf{G}(t_i,\ \mathcal{U}_i) - \mathbf{G}(t_{i-1}, \mathcal{U}_{i-1}) := \mathbf{G}_d(t_i, \mathcal{U}_i),$$

where $t_i = i/n$. Similarly, $\Delta\epsilon_i = H_d(\mathcal{V}_i)$. Then, we can define

$$\mathbf{U}_d(t, \mathcal{I}_i) := \mathbf{G}_d(t,\ \mathcal{U}_i) H_d(\mathcal{V}_i). \tag{31}$$

Lemma 1. *Under Assumptions 1–8,*

$$\mathbf{G}_d(t,\ \mathcal{U}_i) \in Lip_2 \quad \textit{and} \quad \sup_{0\le t\le 1} \|\mathbf{G}_d(t,\ \mathcal{U}_i)\|_4 < \infty.$$

Proof. Note that

$$\begin{aligned}
&\sup_{0\le t_i<t_j\le 1} \frac{\|\mathbf{G}_d(t_j, \mathcal{U}_0) - \mathbf{G}_d(t_i, \mathcal{U}_0)\|_2}{|t_j - t_i|} \\
&\le \sup_{0\le t_i<t_j\le 1} \frac{\|\mathbf{G}(t_j, \mathcal{U}_0) - \mathbf{G}(t_{j-1}, \mathcal{U}_0) - [\mathbf{G}(t_i, \mathcal{U}_0) - \mathbf{G}(t_{i-1}, \mathcal{U}_0)]\|_2}{|t_j - t_i|} \\
&\le \sup_{0\le t_i<t_j\le 1} \frac{\|\mathbf{G}(t_j, \mathcal{U}_0) - \mathbf{G}(t_i, \mathcal{U}_0)\|_2}{|t_j - t_i|} + \sup_{0\le t_{i-1}<t_{j-1}\le 1} \frac{\|\mathbf{G}(t_{j-1}, \mathcal{U}_0) - \mathbf{G}(t_{i-1}, \mathcal{U}_0)\|_2}{|t_{j-1} - t_{i-1}|},
\end{aligned}$$

where $t_j - t_i = t_{j-1} - t_{i-1}$ by definition. By Assumption 2,

$$\sup_{0\le t_i<t_j\le 1} \frac{\|\mathbf{G}(t_j, \mathcal{U}_0) - \mathbf{G}(t_i, \mathcal{U}_0)\|_2}{|t_j - t_i|} < \infty$$

and

$$\sup_{0\le t_{i-1}<t_{j-1}\le 1} \frac{\|\mathbf{G}(t_{j-1}, \mathcal{U}_0) - \mathbf{G}(t_{i-1}, \mathcal{U}_0)\|_2}{|t_{j-1} - t_{i-1}|} < \infty$$

Thus,

$$\sup_{0\le t_i<t_j\le 1} \frac{\|\mathbf{G}_d(t_j, \mathcal{U}_0) - \mathbf{G}_d(t_i, \mathcal{U}_0)\|_2}{|t_j - t_i|} < \infty,$$

which means $\mathbf{G}_d(t,\ \mathcal{U}_i) \in \mathrm{Lip}_2$. Moreover, by Assumption 2,

$$\sup_{0\le t_i\le 1} \|\mathbf{G}_d(t_i,\ \mathcal{U}_i)\|_4 \le \sup_{0\le t_i\le 1} \|\mathbf{G}(t_i,\ \mathcal{U}_i)\|_4 + \sup_{0\le t_{i-1}\le 1} \|\mathbf{G}(t_{i-1},\ \mathcal{U}_{i-1})\|_4 < \infty.$$

Therefore, the lemma easily follows.

Lemma 2. *Under Assumptions 1–8,*

$$\left|\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\Delta\mathbf{x}_i\Delta\epsilon_i\right| = O_{\mathbb{P}}(1)$$

Proof. Let $\psi_d = \frac{1}{n}\sum_{i=1}^{n}\Delta\mathbf{x}_i\Delta\epsilon_i$. Define

$$\psi_{d,k} = \frac{1}{n}\sum_{i=1}^{n}\mathcal{P}_{i-k}\Big(\Delta\mathbf{x}_i\Delta\epsilon_i\Big), \tag{32}$$

where the projection operator is defined by $\mathcal{P}_k(\cdot) := \mathbb{E}[\cdot|\mathcal{I}_k] - \mathbb{E}[\cdot|\mathcal{I}_{k-1}]$ with $\mathcal{I}_k = (\cdots, \zeta_{k-2}, \zeta_{k-1}, \zeta_k)$ introduced by Assumption 5. Then,

$$\begin{aligned}
\|\psi_{d,k}\|^2 &= \left\|\frac{1}{n}\sum_{i=1}^{n}\mathcal{P}_{i-k}\Big(\Delta\mathbf{x}_i\Delta\epsilon_i\Big)\right\|^2 \\
&\le \frac{1}{n^2}\sum_{i=1}^{n}\|\mathbb{E}(\Delta\mathbf{x}_i\Delta\epsilon_i|\mathcal{I}_{i-k}) - \mathbb{E}(\Delta\mathbf{x}_i\Delta\epsilon_i|\mathcal{I}_{i-k-1})\|^2 \\
&= \frac{1}{n^2}\sum_{i=1}^{n}\left\|\mathbb{E}(\Delta\mathbf{x}_i\Delta\epsilon_i|\mathcal{I}_{i-k}) - \mathbb{E}(\Delta\mathbf{x}^*_{i,k}\Delta\epsilon^*_{i,k}|\mathcal{I}_{i-k})\right\|^2 \\
&\le \frac{1}{n^2}\sum_{i=1}^{n}\left\|\Delta\mathbf{x}_i\Delta\epsilon_i - \Delta\mathbf{x}^*_{i,k}\Delta\epsilon^*_{i,k}\right\|^2,
\end{aligned} \tag{33}$$

where $\Delta\mathbf{x}^*_{i,k}\Delta\epsilon^*_{i,k}$ denotes $\Delta\mathbf{x}_i\Delta\epsilon_i$ with ζ_{i-k} replaced by its IID copy ζ^*_{i-k}. In the second equality, $\mathbb{E}(\Delta\mathbf{x}_i\Delta\epsilon_i|\mathcal{I}_{i-k-1})$ can be replaced by $\mathbb{E}(\Delta\mathbf{x}^*_{i,k}\Delta\epsilon^*_{i,k}|\mathcal{I}_{i-k})$ because $\Delta\mathbf{x}^*_{i,k}\Delta\epsilon^*_{i,k}$ is independent of ζ_{i-k}. The last inequality in (33) is due to Jensen's Inequality. Thus, by (33), we have

$$\|\psi_{d,k}\|^2 \le \frac{1}{n^2}\sum_{i=1}^{n}\delta_2^2(\mathbf{U}_d, k) = \frac{1}{n}\delta_2^2(\mathbf{U}_d, k), \tag{34}$$

where $\delta_2(\mathbf{U}_d, k)$ is the physical dependence measure (2) for $\mathbf{U}_d(t, \mathcal{I}_i)$ introduced by (31). Moreover, by (32) and by the definition of projection operator,

$$\sum_{k=0}^{\infty}\psi_{d,k} = \frac{1}{n}\sum_{i=1}^{n}\sum_{k=0}^{\infty}\mathcal{P}_{i-k}\Big(\Delta\mathbf{x}_i\Delta\epsilon_i\Big) = \frac{1}{n}\sum_{i=1}^{n}\Delta\mathbf{x}_i\Delta\epsilon_i. \tag{35}$$

Then, by (35) and $\mathbb{E}\psi_d = 0$,

$$\psi_d - \mathbb{E}\psi_d = \sum_{k=0}^{\infty}\psi_{d,k} \tag{36}$$

Thus, by (34), (36), and Assumption 5,

$$\|\psi_d - \mathbb{E}\psi_d\| \le \sum_{k=0}^{\infty} \|\psi_{d,k}\| \le \frac{1}{\sqrt{n}} \sum_{k=0}^{\infty} \delta_2(\mathbf{U}_d, k) = O\left(\frac{1}{\sqrt{n}}\right). \tag{37}$$

By $\psi_d = \frac{1}{n}\sum_{i=1}^n \Delta\mathbf{x}_i \Delta\epsilon_i$ and $\mathbb{E}\psi_d = 0$, (37) leads to

$$\left\| \frac{1}{n} \sum_{i=1}^{n} \Delta\mathbf{x}_i \Delta\epsilon_i \right\| = O\left(\frac{1}{\sqrt{n}}\right).$$

Thus, the lemma follows.

Lemma 3. *Under Assumptions 1–8,*

$$|\hat{\boldsymbol{\beta}}_D - \boldsymbol{\beta}| = O_{\mathbb{P}}\left(\frac{1}{\sqrt{n}}\right), \tag{38}$$

where $\hat{\boldsymbol{\beta}}_D$ is the differencing estimator of $\boldsymbol{\beta}$ in (11).

Proof. Recall the differencing estimator $\hat{\boldsymbol{\beta}}_D$ given by (11). From (9),

$$\begin{aligned}
\hat{\boldsymbol{\beta}}_D &= \left(\sum_{i=1}^{n} \Delta\mathbf{x}_i \Delta\mathbf{x}_i^\top\right)^{-1} \sum_{i=1}^{n} \Delta\mathbf{x}_i \Delta y_i \\
&= \left(\sum_{i=1}^{n} \Delta\mathbf{x}_i \Delta\mathbf{x}_i^\top\right)^{-1} \sum_{i=1}^{n} \Delta\mathbf{x}_i \left(\Delta\mathbf{x}_i^\top \boldsymbol{\beta} + O(1/n) + \Delta\epsilon_i\right),
\end{aligned}$$

which leads to

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_D - \boldsymbol{\beta}) = \left(\frac{1}{n}\sum_{i=1}^{n} \Delta\mathbf{x}_i \Delta\mathbf{x}_i^\top\right)^{-1} \frac{1}{n}\sum_{i=1}^{n} \Delta\mathbf{x}_i O\left(1/\sqrt{n}\right) + \left(\frac{1}{n}\sum_{i=1}^{n} \Delta\mathbf{x}_i \Delta\mathbf{x}_i^\top\right)^{-1} \frac{1}{\sqrt{n}}\sum_{i=1}^{n} \Delta\mathbf{x}_i \Delta\epsilon_i.$$

Since $i/n \in [0,1]$, Assumption 6 ensures that $O(1/n)$ in (9) is independent of i. Hence

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_D - \boldsymbol{\beta}) = O\left(1/\sqrt{n}\right) \left(\frac{1}{n}\sum_{i=1}^{n} \Delta\mathbf{x}_i \Delta\mathbf{x}_i^\top\right)^{-1} \frac{1}{n}\sum_{i=1}^{n} \Delta\mathbf{x}_i + \left(\frac{1}{n}\sum_{i=1}^{n} \Delta\mathbf{x}_i \Delta\mathbf{x}_i^\top\right)^{-1} \frac{1}{\sqrt{n}}\sum_{i=1}^{n} \Delta\mathbf{x}_i \Delta\epsilon_i \tag{39}$$

Moreover, by Lemma 1, $\Delta\mathbf{x}_i$ is *locally stationary*. Thus, for all $i = 1, \ldots, n$,

$$|\Delta\mathbf{x}_i| = O_{\mathbb{P}}(1),$$

which also leads to

$$\left| \frac{1}{n} \sum_{i=1}^{n} \Delta \mathbf{x}_i \right| = O_{\mathbb{P}}(1). \tag{40}$$

Furthermore, by Assumption 3,

$$\left| \frac{1}{n} \sum_{i=1}^{n} \Delta \mathbf{x}_i \Delta \mathbf{x}_i^{\top} \right|_M^{-1} = O_{\mathbb{P}}(1), \tag{41}$$

where $|\cdot|_M$ is a *matrix norm* induced by the Euclidean norm on a vector that we introduced in Section 2. By applying (40), (41), and Lemma 2 to (39), the lemma follows.

Remark 8. (Limiting Distribution of $\hat{\boldsymbol{\beta}}_D$). Recall that, by Assumption 3 and (31), $\Delta \mathbf{x}_i = \mathbf{G}_d(i/n,\ \mathcal{U}_i)$ and $\Delta \mathbf{x}_i \Delta \epsilon_i = \mathbf{U}_d(i/n, \mathcal{F}_i)$, respectively. Under Assumptions 1–5, by the m-dependence approximation in Proof of Theorem 3.1 of Zhang and Wu (2012),

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \Delta \mathbf{x}_i \Delta \epsilon_i \Rightarrow N\left(0,\ \Lambda\right), \tag{42}$$

where $\Lambda := \int_0^1 \sum_{k \in \mathbb{Z}} \mathbb{E}\left[\mathbf{U}_d(t, \mathcal{F}_0) \cdot \mathbf{U}_d(t, \mathcal{F}_k)^{\top}\right] dt$. Here $\Rightarrow$ denotes a convergence in distribution. In addition, $\left(\frac{1}{n} \sum_{i=1}^{n} \Delta \mathbf{x}_i \Delta \mathbf{x}_i^{\top}\right)^{-1} \overset{\mathbb{P}}{\to} \Omega := \left(\int_0^1 M_d(t) dt\right)^{-1}$, where $M_d(t)$ is defined by Assumption 3. Then, by (39)–(42) and the Slutsky Theorem,

$$\sqrt{n}\left(\hat{\boldsymbol{\beta}}_D - \boldsymbol{\beta}\right) \Rightarrow N\left(0, \Omega \cdot \Lambda \cdot \Omega^{\top}\right).$$

Lemma 4. *Under Assumptions 1–8,*

$$\sup_{0 \le t \le 1} \left| \sum_{i=1}^{n} w_n(t,i)(\epsilon_i - \sigma_\epsilon Z_i) \right| = o_{\mathbb{AS}}\left(\frac{\log(n)}{n^{1-1/q} b_n} \right), \tag{43}$$

where $\sigma_\epsilon^2 = \sum_{j \in \mathbb{Z}} \mathbb{E}(\epsilon_0 \epsilon_j)$ *in (18) and* $q = \min(r, 4)$ *from Assumptions 4 and 8. Here* Z_i *is an IID standard normal random variable.*

Proof. First, let $Z_i = \mathbb{B}(i) - \mathbb{B}(i-1)$, where $\mathbb{B}(\cdot)$ is the standard Brownian motion. Then,

$$
\begin{aligned}
&\sup_{0\le t\le 1}\left|\sum_{i=1}^{n} w_n(t,i)\epsilon_i - \sigma_\epsilon \sum_{i=1}^{n} w_n(t,i)Z_i\right| \\
&= \sup_{0\le t\le 1}\left|w_n(t,n)\sum_{j=1}^{n}(\epsilon_j - \sigma_\epsilon Z_j) - \sum_{j=2}^{n}[w_n(t,j) - w_n(t,j-1)]\sum_{i=1}^{j}(\epsilon_i - \sigma_\epsilon Z_i)\right| \\
&\le \sup_{0\le t\le 1}\left\{\left|w_n(t,n)\right|\left|\sum_{j=1}^{n}(\epsilon_j - \sigma_\epsilon Z_j)\right| + \sum_{j=2}^{n}\left|w_n(t,j) - w_n(t,j-1)\right|\left|\sum_{i=1}^{j}(\epsilon_i - \sigma_\epsilon Z_i)\right|\right\} \\
&\le \sup_{0\le t\le 1}\left\{\max_{j\le n}\left|\sum_{i=1}^{j}\epsilon_i - \sigma_\epsilon \mathbb{B}(j)\right| \left[|w_n(t,n)| + \sum_{j=2}^{n}|w_n(t,j) - w_n(t,j-1)|\right]\right\} \\
&\le \max_{i\le n}\left|\sum_{k=1}^{i}\epsilon_k - \sigma_\epsilon \mathbb{B}(i)\right| \sup_{0\le t\le 1}\left[|w_n(t,n)| + \sum_{j=2}^{n}|w_n(t,j) - w_n(t,j-1)|\right], \qquad (44)
\end{aligned}
$$

where the first equality is due to the *summation-by-parts* formula. By Assumption 4, error ϵ_i is a mean-zero and stationary process with $\mathbb{E}|\epsilon_i|^r < \infty$, $r > 3$. Under the condition (6) of Assumption 4, there is a richer probability space such that the following strong invariance principle holds true Wu (2007):

$$
\max_{i\le n}\left|\sum_{k=1}^{i}\epsilon_k - \sigma_\epsilon \mathbb{B}(i)\right| = o_{\mathbb{AS}}\left(n^{1/q}\log(n)\right), \qquad (45)
$$

where $q = \min(r, 4)$. Moreover, by using that the kernel $K(\cdot)$ in the local-linear weight $w_n(t,i)$ is bounded and Lipschitz-continuous under Assumption 7,

$$
\sup_{0\le t\le 1}\left[|w_n(t,n)| + \sum_{j=2}^{n}|w_n(t,j) - w_n(t,j-1)|\right] = O\left(\frac{1}{nb_n}\right). \qquad (46)
$$

Then, by applying (45) and (46) to (44), we have

$$
\sup_{0\le t\le 1}\left|\sum_{i=1}^{n} w_n(t,i)\epsilon_i - \sigma_\epsilon \sum_{i=1}^{n} w_n(t,i)Z_i\right| = o_{\mathbb{AS}}\left(\frac{n^{1/q}\log(n)}{nb_n}\right),
$$

which leads to the lemma.

Lemma 5. *Let $k_n \asymp n^{1/3}$. Then, under Assumptions 1–8,*

$$\hat{\sigma}_\epsilon^2 = \sigma_\epsilon^2 + O_{\mathbb{P}}(n^{-1/3}), \tag{47}$$

where $\hat{\sigma}_\epsilon^2$ is the subseries variance estimate of σ_ϵ^2 introduced by (21).

Proof. For notational convenience, we let $y_i^* = y_i - \mathbf{x}_i^\top \boldsymbol{\beta}$. Then, a simple computation shows that we can rewrite $\hat{\sigma}_\epsilon^2$ as follows

$$\begin{aligned}\hat{\sigma}_\epsilon^2 = {} & \frac{1}{2(m-1)k_n} \sum_{i=1}^{m-1} \left[\sum_{j=1}^{k_n} \left(y^*_{j+ik_n} - y^*_{j+(i-1)k_n} \right) \right]^2 \\ & + \frac{1}{2(m-1)k_n} \sum_{i=1}^{m-1} \left[\sum_{j=1}^{k_n} \left(\mathbf{x}_{j+ik_n} - \mathbf{x}_{j+(i-1)k_n} \right)^\top (\hat{\boldsymbol{\beta}}_D - \boldsymbol{\beta}) \right]^2 \\ & - \frac{1}{(m-1)k_n} \sum_{i=1}^{m-1} \left[\sum_{j=1}^{k_n} \left(y^*_{j+ik_n} - y^*_{j+(i-1)k_n} \right) \sum_{j=1}^{k_n} \left(\mathbf{x}_{j+ik_n} - \mathbf{x}_{j+(i-1)k_n} \right)^\top (\hat{\boldsymbol{\beta}}_D - \boldsymbol{\beta}) \right]. \end{aligned} \tag{48}$$

By Carlstein (1986) and Wu and Zhao (2007), we have

$$\frac{1}{2(m-1)k_n} \sum_{i=1}^{m-1} \left[\sum_{j=1}^{k_n} \left(y^*_{j+ik_n} - y^*_{j+(i-1)k_n} \right) \right]^2 = \sigma_\epsilon^2 + O_{\mathbb{P}}(n^{-1/3}). \tag{49}$$

Moreover, by $k_n \asymp n^{1/3}$, Assumption 2 and Lemma 3, we have

$$\frac{1}{2(m-1)k_n} \sum_{i=1}^{m-1} \left[\sum_{j=1}^{k_n} \left(\mathbf{x}_{j+ik_n} - \mathbf{x}_{j+(i-1)k_n} \right)^\top (\hat{\boldsymbol{\beta}}_D - \boldsymbol{\beta}) \right]^2 = O_{\mathbb{P}}(n^{-2/3}). \tag{50}$$

Furthermore, by (49), (50), and the Cauchy-Schwarz inequality,

$$\frac{1}{(m-1)k_n} \sum_{i=1}^{m-1} \left[\sum_{j=1}^{k_n} \left(y^*_{j+ik_n} - y^*_{j+(i-1)k_n} \right) \sum_{j=1}^{k_n} \left(\mathbf{x}_{j+ik_n} - \mathbf{x}_{j+(i-1)k_n} \right)^\top (\hat{\boldsymbol{\beta}}_D - \boldsymbol{\beta}) \right] = O_{\mathbb{P}}(n^{-1/3}). \tag{51}$$

Thus, by applying (49)–(51) to (48), the lemma follows easily.

7.1. Proof of Theorem 1

Recall the two-stage local linear regression estimator in (15). By (1) and by Taylor's expansion of $\mu(\cdot)$, we show

$$\begin{aligned}
\tilde{\mu}(t) &= \sum_{i=1}^{n} w_n(t,i)(y_i - \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}_D) \\
&= \sum_{i=1}^{n} w_n(t,i)\mathbf{x}_i^\top(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}_D) + \sum_{i=1}^{n} w_n(t,i)(\mu(i/n) + \epsilon_i) \\
&= \sum_{i=1}^{n} w_n(t,i)\mathbf{x}_i^\top(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}_D) + \sum_{i=1}^{n} w_n(t,i)\left[\mu(t) + (i/n - t)\mu'(t) + (i/n - t)^2\mu''(t_0)/2 + \epsilon_i\right],
\end{aligned}$$

where $t_0 \in (t,\ i/n)$. Thus, by $\sum_{i=1}^{n} w_n(t,i) = 1$ and by $\sum_{i=1}^{n}(t - i/n)w_n(t,i) = 0$ from the first-order conditions of the local-linear regression,

$$\tilde{\mu}(t) - \mu(t) = \sum_{i=1}^{n} w_n(t,i)\mathbf{x}_i^\top(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}_D) + \sum_{i=1}^{n} w_n(t,i)(i/n - t)^2\mu''(t_0)/2 + \sum_{i=1}^{n} w_n(t,i)\epsilon_i. \quad (52)$$

By Assumption 2 and Lemma 3, for any fixed $t \in (0,1)$,

$$\left|\sum_{i=1}^{n} w_n(t,i)\mathbf{x}_i^\top(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}_D)\right| = O_{\mathbb{P}}\left(n^{-1/2}\right). \quad (53)$$

Moreover, by Assumptions 6–8,

$$\left|\sum_{i=1}^{n} w_n(t,i)(i/n - t)^2\mu''(t_0)\right| = O\left(b_n^2\right), \quad (54)$$

for any fixed $t \in (0,1)$. Furthermore, by Assumption 4,

$$\left|\sum_{i=1}^{n} w_n(t,i)\epsilon_i\right| = O_{\mathbb{P}}\left(\frac{1}{\sqrt{nb_n}}\right), \quad (55)$$

for any fixed $t \in (0,1)$. Therefore, by applying (53)–(55) to (52), the theorem follows.

7.2. Proof of Theorem 2

From Eq. (1), we have $y_i - \mathbf{x}_i^\top \boldsymbol{\beta} = \mu(i/n) + \epsilon_i$, where ϵ_i satisfies Assumption 4. Since $\mathbf{x}_i$ is local-stationary by Assumption 2, we have

$$\sup_{0\le t\le 1}\left|\sum_{i=1}^{n} w_n(t,i)\mathbf{x}_i^\top\right| = O_{\mathbb{P}}(1). \tag{56}$$

Recall $Z_i = \mathbb{B}(i) - \mathbb{B}(i-1)$, where $\mathbb{B}(\cdot)$ is the standard Brownian motion. By (1) and by a Taylor's expansion on $\mu(\cdot)$,

$$\begin{aligned}
&\sup_{0\le t\le 1}\left|\sum_{i=1}^{n} w_n(t,i)[y_i - \mathbf{x}_i^\top\hat{\boldsymbol{\beta}}_D - \mu(t)] - \sigma_\epsilon\sum_{i=1}^{n} w_n(t,i)Z_i\right| \\
&\le \sup_{0\le t\le 1}\left|\sum_{i=1}^{n} w_n(t,i)\mathbf{x}_i^\top(\boldsymbol{\beta}-\hat{\boldsymbol{\beta}}_D) + \sum_{i=1}^{n} w_n(t,i)\epsilon_i - \sigma_\epsilon\sum_{i=1}^{n} w_n(t,i)Z_i\right. \\
&\quad \left. + \sum_{i=1}^{n} w_n(t,i)\left[(t-i/n)\mu'(t) + (t-i/n)^2\mu''(t_0)/2\right]\right| \\
&\le \sup_{0\le t\le 1}\left|\sum_{i=1}^{n} w_n(t,i)\mathbf{x}_i^\top(\boldsymbol{\beta}-\hat{\boldsymbol{\beta}}_D)\right| + \sup_{0\le t\le 1}\left|\sum_{i=1}^{n} w_n(t,i)\epsilon_i - \sigma_\epsilon\sum_{i=1}^{n} w_n(t,i)Z_i\right| \\
&\quad + (1/2)\sup_{0\le t\le 1}\left|\sum_{i=1}^{n}(t-i/n)^2 w_n(t,i)\mu''(t_0)\right| \\
&\le \left|\boldsymbol{\beta}-\hat{\boldsymbol{\beta}}_D\right|\sup_{0\le t\le 1}\left|\sum_{i=1}^{n} w_n(t,i)\mathbf{x}_i^\top\right| + \sup_{0\le t\le 1}\left|\sum_{i=1}^{n} w_n(t,i)\epsilon_i - \sigma_\epsilon\sum_{i=1}^{n} w_n(t,i)Z_i\right| \\
&\quad + (1/2)\sup_{0\le t\le 1}\left|\sum_{i=1}^{n}(t-i/n)^2 w_n(t,i)|\mu''(t_0)|\right| \\
&= O_{\mathbb{P}}(n^{-1/2}) + o_{\mathbb{AS}}\left(\frac{\log(n)}{n^{1-1/q}b_n}\right) + O(b_n^2) \\
&= O_{\mathbb{P}}\left(n^{-1/2} + \frac{\log(n)}{n^{1-1/q}b_n} + b_n^2\right),
\end{aligned} \tag{57}$$

where $t_0 \in (t,\ i/n)$. The first inequality is due to the Taylor's expansion on $\mu(\cdot)$. The second inequality uses the fact $\sum_{i=1}^{n}(t-i/n)w_n(t,i) = 0$ from the first-order conditions of the local-linear regression. The second last equality follows from (56) and Lemmas 3 and 4. In addition, we utilize Assumptions 6–8 and the fact that the weight $w_n(t,i)$ in (14) involves the kernel $K\left(\frac{t-i/n}{b_n}\right)$ with a compact support $[-1,\ 1]$. Hence, by applying (15) to

(57), we have

$$\sqrt{nb_n} \sup_{0 \le t \le 1} \left| \tilde{\mu}(t) - \mu(t) - \sigma_\epsilon \sum_{i=1}^{n} w_n(t,i) Z_i \right| = O_{\mathbb{P}} \left(b_n^{1/2} + \frac{\log(n)}{n^{1/2-1/q} b_n^{1/2}} + n^{1/2} b_n^{5/2} \right).$$

Therefore, by Assumption 8, the theorem follows.

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